MIL-S-20556
4 January 1952

MILITARY SPECIFICATION

SHELL, STEEL, FOR SMOOTH-BORE MORTARS, MANUFACTURED FROM STEEL CASTINGS

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specification:

Army 50-37-3
23 May 1945

This specification consists of this cover sheet, Specification 50-37-3, dated 23 May 1945, and Amendment 2, dated 7 February 1949, attached hereto, without modification.

When a request for this specification is received by a supplying activity it will be necessary to attach this cover sheet to the pertinent specification before issue.

Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.

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U. S. ARMY
SPECIFICATION

No. 50-37-3
23 May 1945

SHELL, STEEL, FOR SMOOTH-BORE MORTARS, MANUFACTURED FROM STEEL CASTINGS

A. APPLICABLE SPECIFICATIONS.

A-1. The Ordnance Department drawings pertaining to the shell under contract and all drawings and specifications listed or referred to thereon form a part of this specification.

A-2. All drawings and specifications shall be of the issue in effect on the date of invitations for bids.

B. TYPE.

B-1. This specification covers high explosive and practice shell for smooth-bore mortars, manufactured from steel castings.

C. MATERIAL & WORKMANSHIP.

C-1. Material and parts shall be in accordance with applicable specifications and drawings.

C-2. The shell shall be machined from steel castings produced by the open hearth or electric furnace process and in accordance with the applicable provisions of this specification.

C-3. All shell shall be sound and free from cracks, porosity, and excessive surface imperfections. (See Par F-3 & F-4).

C-4. The interior of all shell shall be free from burrs, scale, fins, sharp edges, and excessive surface imperfections. Sand cast finish is acceptable. (See Par F-5).

D. GENERAL REQUIREMENTS.

D-1. All castings shall be heat treated. The contractor shall demonstrate by manufacturing a process qualification lot of castings that the composition of steel and heat treatment will produce sound castings of uniform quality, having the physical properties and complying with the fragmentation requirements herein specified. This heat treatment, once established, will not be changed unless periodic test samples indicate failure to meet physical requirements.

D-2. The shell cavity shall be in finished form in the as-cast condition. No machining of the interior, except the fuze hole, will be permitted except as hereinafter authorized in Par F-5.

E. DETAIL REQUIREMENTS.

E-1. The shell shall meet all requirements prescribed in drawings and applicable specifications.

E-2. Chemical Composition:- The composition of the shell steel shall meet the following chemical requirements:

		60 m/m, M49A2	81 m/m, M43A1 & M43A1B1
Carbon	Percent	.40 max.	.45 max.
Silicon	"	.15 min. to .40 max.	.15 min. to .40 max.
Manganese	"	.80 max.	.80 max.
Phosphorous	"	.06 max.	.06 max.
Sulphur	"	.075 max.	.075 max.

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Other chemical compositions will be permitted only upon specific approval of the Chief of Ordnance.

E-3. Physical Properties:- The physical properties as determined from test bars shall meet the following requirements:

Yield Strength	35,000 lbs. per sq. in. min.
Elongation	15%, min.
Reduction of area	25%, min.

E-4. Finished shell shall withstand a minimum internal air pressure of 150 lbs. per sq. inch without evidence of leaking.

E-5. Fragmentation Properties:- When fragmented in the pit, no shell shall yield less than 80% recovered metal.

E-6. Cleaning Prior to Painting:- Upon completion of the machining operations, the shell shall be cleaned to free them of all grease, dirt, chips, and other foreign material, preparatory to painting. The shell surfaces to be painted shall be cleaned by one of the methods covered by Specification AXS-1245. These cleaned surfaces shall not be recontaminated prior to painting. At the time of painting these surfaces shall meet the various performance requirements of Specification AXS-1245. The interior cavity of each shell shall be cleaned by the shot or sand blast method, or by any other method approved by the Chief of Ordnance.

E-7. Coating:- The exterior coating and the cavity coating specified on the drawing for the shell shall be applied in such a manner as to produce a tightly adherent coating of uniform thickness over the entire specified areas. The shell shall be stencilled or marked as prescribed on the drawing. All paint and marking shall be dry to the touch before shells are packed.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F-1. Inspection:- Inspection shall be as prescribed in U. S. Army Specification No. 50-0-1, "General Specifications for Ammunition Except Small Arms Ammunition", and Federal Specification No. QQ-M-151, "Metals: General Specification for Inspection of".

F-2. Lots:-

F-2a. The maximum size of lots shall be 20,000. All shells in a lot shall be from castings made by the same melting and the same heat treating process. More than one heat of steel meeting the chemical and physical requirements may be used in one lot of castings.

F-2b. A smaller quantity may constitute a lot at the discretion of the inspector.

F-2c. The first lot manufactured on any contract shall consist of 2500 ± 500 shell, except when a new contract immediately follows the manufacture of satisfactory material of the same character.

F-3. Porosity Test:- Each completely machined shell, after having been thoroughly cleaned, but prior to applying interior and exterior coatings, shall be subjected to a minimum internal air pressure of 150 lbs. per sq. in. for a minimum time of 15 seconds. For this test, the shell shall be completely immersed in a clear water solution which may contain a suitable rust inhibitor. The presence of bubbles either on or leaving the surface on any portion of the shell shall be regarded as evidence of porosity and the shell rejected. Because of the importance of this test in proving soundness of shell, each shell, after having passed the test shall be stamped immediately with an identifying mark. (See Fig. I).

F-4. Exterior of Shell:- The exterior surfaces of each shell shall be carefully inspected after machining for casting defects. Samples of shell showing the limits of casting defects of a surface nature which are acceptable

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without repairing shall be furnished or approved by the Chief of Ordnance. Samples of shell showing casting defects of a surface nature which are beyond the above sample limits but which need not be repaired by welding shall be furnished or approved by the Chief of Ordnance. All such shell shall be repaired in accordance with the provisions of Par F-9a in order to be acceptable. All shell with casting defects beyond the limits established by the above samples shall be repaired by welding in accordance with the provisions of Par F-9b in order to be acceptable. Surface imperfections in the bourrelet areaconnecting two or more grooves, and imperfections in the nose threads extending beyond the crests of two threads or occurring in the first full thread shall be rejected.

F-5. Cavity of Shell.- The interior of each shell shall be inspected for the presence of burrs, scale, sharp projections, fins, and sharp edges. All shell having these defects will be rejected but may be repaired. Samples showing the limit of acceptable surface imperfections shall be furnished or approved by the Chief of Ordnance. Shell cavities may be repaired to bring them into comformance with applicable drawings and specifications. For correction of cavity wall, Methods a and b as stated below shall be used and, for the bottom of the cavity, Method c shall be used. In any case, reworked shell shall conform to all the dimensions and requirements of the applicable shell drawings and specifications. This reworking may be performed either before or after shot or sand blasting, but reworking in accordance with Methods a and c shall be followed by a shot or sand blasting operation.

Method a. The operation shall be performed by holding the shell on the outside diameter and removing from the cavity wall only that metal in excess of the permissible run-out with a single point tool or equivalent which follows a cam, the outline of which produces the contour of the cavity as specified on the drawing, with a smooth blending at all points eliminating shoulders or sharp edges of any kind and with a finish equivalent to the outside machined surface of the shell. In each case the cutting tool shall advance beyond the point where the cutting of metal ceases.

Method b. Hand grinding may be performed for removing such surface imperfections as slight pits, local protrusions, etc., providing the periphery of the ground areas blends with the adjacent area of the cavity wall without shoulders or sharp edges of any kind and with a finish equivalent to the cast surface.

Method c. Spading or rose reaming of the bottom of the cavity may be performed for removing such imperfections as small protrusions, pits, etc., providing the periphery of the altered surface blends with the contour of the cavity without shoulders or sharp edges of any kind and with a finish equivalent to the outside machined surface of the shell.

The cavity of each shell reworked according to any of the above processes shall be inspected 100% by the Inspector.

F-6. Chemical Composition:- Samples for chemical analysis shall be taken from each heat or melt. Failure to comply with the composition shall reject all castings from the heat or melt.

F-7. Physical Properties:- Each heat or melt of castings will have cast with it test bars. These test bars shall be heat treated under the same conditions as the castings which they represent. Normally one test bar shall

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be pulled and its failure to comply with the required minimum physical properties shall reject the castings from this heat or melt. Test bars having casting imperfections may be replaced. If the test bar representing a heat or melt of castings fails to conform to the physical requirements, two additional test bars from the heat or melt that failed may be selected by the Inspector to represent the heat or melt and shall be tested. In case of failure of either test bar on retest, the heat or melt of castings shall be rejected, subject to reheat treatment.

F-8. Reheat Treatment:- A heat or melt of castings which does not meet the required physical properties may be reheat treated once before final rejection. If upon reheat treatment, the physical properties are above the minimum requirements, castings from this heat or melt will be accepted.

F-9. Repairs of External Surface Imperfections:-

F-9a. Shell having casting defects of a surface nature within the limits established by the approved samples referred to in Par F-4 may be refinished by any method approved by the Chief of Ordnance. Such refinishing operations shall improve the character of the defective surfaces by blending such surfaces with the contours of the shell as prescribed by the drawings.

F-9b. Casting defects which are beyond the limits established by the approved samples referred to in Par F-4 may be repaired by using either arc welding (shielded arc) or gas welding methods. All welded shell shall be stress-relieved by subjecting them to a temperature range of 1000°F. to 1200°F. for a minimum period of one-half hour or by any other method approved by the Chief of Ordnance, and then resubjected to the porosity test. Failure to meet the porosity test after such repairing and stress-relieving shall finally reject the shell.

F-9c. Shell which failed to meet the porosity test (See Par F-3) may be repaired by welding by using either arc welding (shielded arc) or gas welding methods only. Shell which have been aged for a period of time in an attempt to promote sealing of leaks are not acceptable. All welded shell shall be stress-relieved by subjecting them to a temperature ranging from 1000°F. to 1200°F., for a minimum period of one-half hour or by any other method approved by the Chief of Ordnance, and then resubjected to the porosity test. Failure to meet the porosity test after such repairing and stress-relieving shall finally reject the shell.

F-9d. All shell which have been repaired and accepted under the provisions of Par F-9b and F-9c shall be placed only in miscellaneous lots not exceeding 5000 shell. Periodic tests shall be made by the Inspector to insure that the required physical properties have been met. Repaired and stress-relieved shell may be heat treated only once in order to meet the physical requirements. Special tests for soundness and strength may be conducted at any time by the Chief of Ordnance on repaired shell.

F-10. Test for Physical Properties of Shell Repaired by Welding.-

F-10a. At the option of the contractor, the yield strength of the repaired shell may be determined by a hydrostatic test in place of determining yield strength from tension test specimens.

F-10b. For the hydrostatic test, twenty (20) shell shall be selected from each miscellaneous shell lot by the Inspector as prepared in accordance with Par. F-9d. The shell shall be subjected to a minimum internal hydrostatic pressure of 6,500 lbs. per sq. in. in the presence of the inspector followed by a careful inspection by the inspector to ascertain if any distortion has taken place in the shell body.

F-10c. Distortion shall be measured at a distance from the base of the shell equivalent to 40 percent of the distance between the base of the shell and the rear edge of the bourrelet. Distortion shall be defined as an increase in

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diameter produced by the hydrostatic test when measured at two positions on the shell 90° apart, which exceeds .005 in. at either of these positions after the load has been removed. The contractor shall incorporate suitable flat areas on the shell body at the positions where the measurements are to be taken. Approval of these shell by the inspector after passing the hydrostatic test shall be considered as approval of the represented lot.

F-10d. Failure of any one of the test shell in the hydrostatic test, as indicated by distortion beyond the limit specified in Par 10c, shall require the determination of the physical properties of the shell by means of tension tests. Failure of the shell in the tension tests shall reject the lot.

F-11. Fragmentation Test:- Not less than ten shell selected from sound shell representing the first lot shall be fragmented in a recovery pit. Failure of any one shell to yield not less than 80% metal recovered by the Standard Screen, shall reject the lot. The contractor shall, after changing composition and/or heat treatment, prepare and submit another first lot for process qualification. The inspector may at any time during the contract select shell for fragmentation tests upon approval from the Chief of Ordnance. A fragmentation test will not necessarily be required when a new contract immediately follows the manufacture of satisfactory material of the same character.

F-12. The Inspector shall assure himself that the operation of cleaning the cavity and exterior of the shell is performed in such a manner as to remove all foreign material and to insure that the surfaces meet the various performance requirements of Specification AXS-1245.

G. PACKAGING, PACKING, AND MARKING.

G-1. Packing.-

G-1a. Shell shall be packed in commerical type containers so constructed as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

G-1b. Data Card:- Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 lbs. ream weight, shall be prepared for each lot of accepted shell in accordance with form specified below. Two of these cards completely filled out for each lot shall accompany each shipping ticket and six cards completely filled out shall be furnished the Inspector. Lot number of contents shall be marked plainly on each container.

AMMUNITION DATA CARD

Kind:.....Ammunition Lot No.....
Quantity in Lot.....
Dwg. No.....Rev.....Spec. No.....Rev.....
Manufactured by.....Date.....
Contract No.....Order No.....
Packed.....in a box.
Casting Manufacturer.....
Remarks.....
Certified to by.....Inspector.
(Print here name of inspector, district, arsenal, or loading plant, and sub-district, where applicable.)

G-2. Markings.- The marking for shipment shall be in accordance with U. S. Army Standard Specifications for Marking Shipments By Contractor, 100-2.

H. NOTES.

H-1. The shell covered by this specification are for use in Smooth-Bore Mortars.

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H-2. The use of this Specification, whenever applicable, is mandatory on all procuring agencies of the Army.

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NOTE:- Copies of this specification may be obtained from the Office Chief of Ordnance, Army Service Forces, Washington 25, D.C.

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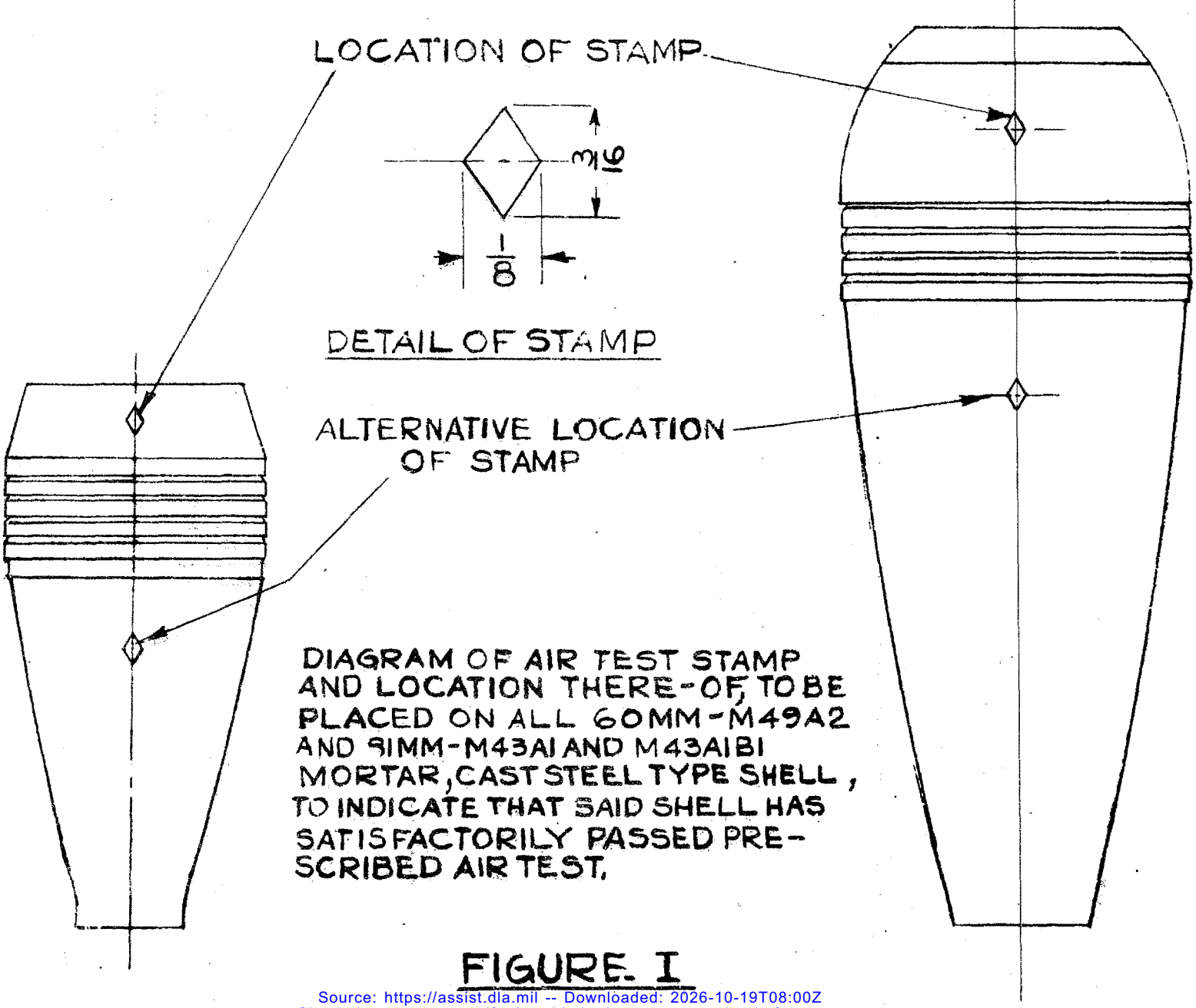


DIAGRAM OF AIR TEST STAMP AND LOCATION THERE-OF, TO BE PLACED ON ALL 60MM-M49A2 AND 91MM-M43A1 AND M43A1B1 MORTAR, CAST STEEL TYPE SHELL, TO INDICATE THAT SAID SHELL HAS SATISFACTORILY PASSED PRE-SCRIBED AIR TEST.

FIGURE. I

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